

# Enigma — Session #11

Quantum Security Expert // Session Briefing

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## Topics Covered:

1. Session History Timeline (10 prior)
2. Roadmap Progress Dashboard
3. Current State & Commercialization
4. Previous Session Analysis (#10)
5. Next Actions — Prioritized
6. Deliverables Inventory

### Project Context

Project: Enigma — Quantum Security Expert Track  
Owner: Jacob Ribbe (Ribbz)  
Phase Status: ALL 8 PHASES COMPLETE + v4.0 Enterprise  
Current Focus: Commercialization & Community Edition  
Codebase: 28,671 LOC // 74 Python files  
Tests: 538 fast-passing (0 failures)  
CLI Flags: 22  
Languages Scanned: 16 (13 AST + 3 IaC)  
Precision: 96.2% // Recall: 100% // F1: 98%  
Standalone: Enigma.exe v4 — 46 MB

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## 1. Session History Timeline

Ten sessions complete (2026-03-21 → 2026-03-27). Track record:

- S1 (03-21) — Genesis: Project setup + Phase 1 research (133-page PDF)
- S2 (03-21) — Phase 1 impl: modular\_math.py 665 LOC / 14 tests
- S3 (03-22) — Marathon: All 8 phases built, 5,501 LOC, Holy Trinity audit
- S4 (03-22) — v2: ML-KEM + Ed25519 + SARIF scanner + pytest migration
- S5 (03-22) — v3: AST scanner + consumer GUI + Sentinel spinoff
- S6 (03-22) — v4 Wave 1-2: Tree-sitter 7 langs + CBOM + PyPI
- S7 (03-23) — v4 Wave 3-5: CNSA + QARS + Auto-fix + Dashboard + VS Code
- S8 (03-23) — Production: 96.2% precision + 7K-file scale test
- S9 (03-26) — 13 AST langs + AI triage + SBOM enrich + call graph
- S10 (03-27) — Multi-lang call graph + team dashboard + direct marketing PDF

## 2. Roadmap Progress Dashboard

All 8 phases COMPLETE. Phase 6 (Migration Scanner) evolved into v4.0 enterprise product.

- Phase 1 — Math Foundations: COMPLETE (665 LOC, 14 tests)
- Phase 2 — Classical Cryptography: COMPLETE (2,063 LOC, 51 tests)
- Phase 3 — Quantum Mechanics: COMPLETE (380 LOC, 13 tests)
- Phase 4 — Quantum Threat Model: COMPLETE (465 LOC, 11 tests)
- Phase 5 — PQC Standards: COMPLETE (~1,000 LOC — LWE, ML-KEM)
- Phase 6 — Migration Scanner: COMPLETE (v4.0, ~15K LOC, 538 tests)
- Phase 7 — QKD + Networking: COMPLETE (371 LOC — BB84, E91, QRNG)
- Phase 8 — Portfolio + GTM: COMPLETE (274 LOC + Marketing Strategy)

Overall: 8/8 phases (100%). Roadmap finished ahead of 28-week plan in 6 days.

## 3. Current State (v4.0 Enterprise)

Commercial tool ready. All 3 remaining limitations resolved in Session 10.

- Codebase: 28,671 LOC across 74 Python files
- Tests: 538 fast-passing / 562 total / 0 failures
- CLI: 22 flags, 6 output formats (text/json/html/sarif/cbom/dashboard)
- Scanner modules: 18+ (tree-sitter, AST, autofix, QARS, dashboard, history, AI triage, SBOM, callgraph)
- Languages: 13 AST  
(Python/Java/Go/JS/TS/Rust/C/C++/C#/Ruby/PHP/Kotlin/Swift/Scala) + 3 IaC
- Risk scoring: 20 NIST/ETSI/BSI-calibrated algorithms + 13 key sizes
- Auto-fix: 36 rules (25 Tier 1 deterministic + 11 Tier 2 template)
- Compliance: CNSA 2.0 + NIST IR 8547 + PCI DSS 4.0 + deadline countdown
- Standalone: Enigma.exe 46 MB + VS Code .vsix 12.59 KB + Docker ready
- Precision: 96.2% / Recall: 100% / F1: 98% / Scale: 7K files in 12s / 32MB RAM

## 4. Commercialization Status

Strategy: sell the TOOL to security consultants (not enterprises directly). User not a PQC expert.

- Pricing: \$749/mo Professional // \$1,499/mo Enterprise
- Best buyers: small PQC consulting firms (5-20 employees)
- Verified buyer list: 7 consulting firms + 5 defense + 8 finance  
(Direct\_Marketing.pdf)
- Revenue projection: \$180K Y1 → \$750K Y2 → \$2.5M Y3 (realistic)

- Market: \$1.15B (2024) → \$7.82B (2030), 37.6% CAGR, NIST 2030 deadline
- Competitive position: #1 of 30 analyzed tools, 0 feature gaps
- Window: now → ~2028 before Snyk/Checkmarx add PQC modules

## 5. Previous Session Recap (Session #10)

Title: Feature Completion + Direct Marketing Intelligence (2026-03-27)

Summary: Holy Trinity pass on 3 limitations (callgraph, reachability, dashboard) — single-pass convergence 21/35 → 30/35. Direct marketing research: 3 parallel agents, 178 tool uses, 500+ web searches producing verified buyer list.

Deliverables completed:

- Multi-language call graph — all 13 tree-sitter languages (was Python-only)
- Tree-sitter AST reachability — replaced regex across all 13 languages
- Team dashboard — scan history, SVG trends, team activity, project comparison
- 4 new CLI flags: --save-history, --history-file, --scanner-name, --team-dashboard
- Enigma.exe v4 rebuilt: 46 MB with all new features
- Direct\_Marketing.pdf: 7 pages, all contacts web-verified

Key learnings:

- Tree-sitter \_extract\_call\_name renamed to avoid Python AST collision
- Best buyer persona: small PQC consulting firms, not enterprise CISOs direct
- DEF CON Demo Labs + Black Hat Arsenal REQUIRE open-source — need community ed
- CNSA 2.0 Jan 2027 deadline = most immediate purchase driver

## 6. Next Actions — Prioritized

From Session #10 close. 9 days elapsed. Conference deadlines tightening.

URGENT (deadline-driven):

- DEF CON 34 Demo Labs — submit by May 1 (need open-source community edition)
- FAA PQC RFI — respond by April 10 to jackson.lemay@faa.gov (5 days)
- Black Hat Arsenal CFP — next window

HIGH (commercialization):

- Build landing page + domain registration
- Publish open-source community edition to GitHub
- Cold outreach: Encryption Consulting, QryptoCyber, evolutionQ (free scan demo)
- Run scanner on 10 popular open-source projects, publish results

MEDIUM (community / credibility):

- Join NIST PQC Forum, OQS Discord, CSA Quantum-Safe WG
- First 10 LinkedIn outreach to defense contractor CISOs
- OWASP AppSec USA CFP submission (mid-2026)

TECH (product polish):

- Re-run precision/recall benchmark with new patterns
- Update Enigma.exe if any modules changed
- Consider: open-source community vs enterprise paid split architecture

## 7. Deliverables Inventory — Full

Scanner modules (18):

- pqc\_scanner.py, treesitter\_scanner.py, autofix.py, readiness.py
- rules\_engine.py, dashboard.py, history.py, sarif\_output.py
- compliance.py, cnsa\_deadlines.py, qars.py, cbom.py
- ai\_triage.py, sbom\_enrichment.py, callgraph.py, reachability.py
- estimator.py, performance.py

Phase modules (from-scratch crypto):

- Phase 1: modular\_math.py
- Phase 2: sha256.py, aes.py, rsa.py, ecc.py, ed25519.py, cryptopals.py, tls\_analyzer.py
- Phase 3: quantum\_circuits.py + 10 Qiskit circuits + teleportation + QFT
- Phase 4: quantum\_threats.py (Shor, Grover, threat assessment)
- Phase 5: pqc\_lattice.py + ml\_kem.py (FIPS 203 all 3 param sets)
- Phase 7: qkd\_protocols.py (BB84, E91, QRNG, hybrid architecture)
- Phase 8: portfolio.py

Product artifacts:

- Enigma.exe (46 MB standalone)
- VS Code extension (pqc-scanner-4.0.0.vsix)
- Docker image + docker-compose.yml
- PyPI package (pyproject.toml ready)
- GitHub Action (action.yml + scan.py)

Documents:

- Quantum\_Security\_Expert\_Roadmap.pdf (original plan)
- Enigma\_Capstone.pdf + PQC\_Executive\_Report.pdf
- Enigma\_PQC\_Scanner\_Corporate\_Overview.pdf (14 pages)
- PQC\_Scanner\_Test\_Guide.pdf + QUICKSTART.md
- MARKETING\_STRATEGY.md + Direct\_Marketing.pdf (verified buyers)
- 10+ research master docs, 40+ agents deployed

## 8. Strategic Notes

- Phase progression on roadmap is LINEAR and COMPLETE — no skipping needed
- Project shifted from study → product → commercialization over 10 sessions
- User is NOT a crypto expert — positioning must reflect this (tool, not advisory)
- Holy Trinity workflow is the confirmed approach (validated across 5 rounds)
- Solo developer = enterprise liability; consulting reseller channel is the play
- Community edition (open-source) required for DEF CON / Black Hat credibility
- 9 days since last session — check for any stale state in research files

## 9. Behavioral Rules Active

- feedback\_holy\_trinity — Use Holy Trinity workflow for all Enigma work
- feedback\_path\_b — Go DEEP on quantum security niche, not broad
- feedback\_no\_public\_release — Actively exploring commercialization (updated)
- feedback\_real\_products — Build REAL solutions, benchmark, no half-assed copies
- feedback\_sell\_to\_consultants — User is NOT crypto expert; sell tool to consultants